



PROGRAMMING WITH PYTHON

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Computer Applications

PROGRAMMING WITH PYTHON

Exception Handling, File I/O, Modules and Packages

File I/O Operations - Excel

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PROGRAMMING WITH PYTHON

Excel

- Openpyxl is a Python library that enables users to read and write Excel files.
- This framework assists in creating functions, formatting spreadsheets, generating reports, and constructing charts directly within Python, eliminating the need to open an Excel application.
- Openpyxl permits users to navigate through worksheets and conduct identical analyses on various data sets simultaneously.



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- ## Install the package

```
: pip install openpyxl
```

Import the package

```
: import openpyxl
```

Loading the workbook

```
: wb = openpyxl.load_workbook('VideoSales.xlsx')
```

```
: type(wb)
```

```
: openpyxl.workbook.workbook.Workbook
```



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Excel

	Rank	Name	Platform	Year	Genre	Publisher	North_Sales	South_Sales	East_Sales	West_Sales
1										
2	1	Cricket Ch	PC	2024	Sports	GameStud	2.15	3.4	1.85	2.6
3	2	Delhi Metr	Mobile	2023	Racing	IndiePlay L	1.2	1.9	0.95	1.3
4	3	Mumbai C	PC	2022	Action	Riverfront	1.75	2.1	1.05	1.6
5	4	Temple Qu	Mobile	2021	Adventure	PixelLot In	1.9	2.5	1.4	1.8
6	5	Royal Ches	PC	2020	Strategy	Banyan Tre	0.95	1.3	0.8	0.9
7	6	Battle of K	Console	2023	Action	Lotus Inter	1.6	2.05	1.1	1.55
8	7	Goa Beach	Mobile	2022	Sports	Coastal Ga	0.75	1.6	0.6	1.1
9	8	Hyderabad	PC	2021	Simulation	Deccan Dig	0.85	1.2	0.7	0.95
10	9	Rajasthan	Console	2020	Racing	SandStorm	0.9	1.1	0.55	0.8
11	10	Bengal My	PC	2024	Puzzle	Ganges Int	1.1	1.5	1.25	1.05
12	11	Karnataka	Mobile	2023	Platformer	Western G	0.7	1.4	0.65	0.9
13	12	Chennai S	Console	2021	Sports	Marina Ga	1.3	1.95	0.9	1.4
14	13	Ganges Bo	Mobile	2020	Racing	Riverfront	0.6	1.1	0.95	0.75
15	14	Space War	PC	2022	Shooter	Orbit Studi	1.4	1.65	0.85	1.2
16	15	Ancient Te	PC	2021	Adventure	Heritage P	0.95	1.05	0.9	0.8
17	16	City Builde	PC	2020	Simulation	UrbanGrid	1.05	1.3	0.7	1
18	17	Kerala Bac	Mobile	2023	Casual	Coconut L	0.55	1.25	0.6	0.85
19	18	Himalayan	PC	2022	Adventure	PeakPoint	0.8	0.95	0.75	0.7
20	19	Auto Ricks	Mobile	2021	Simulation	BusyStreet	1.25	1.8	0.95	1.35
21	20	Street Cric	Mobile	2020	Sports	GullyShot S	1.5	2.2	1.1	1.6
22	21	Indian Rail	PC	2023	Simulation	National L	1.8	2.1	1.35	1.55
23	22	Bollywood	Console	2022	Music	StageLight	1.1	1.85	0.9	1.2
24	23	Startup Ga	PC	2021	Strategy	Silicon Plat	0.85	1.4	0.75	1
25	24	Ancient Fo	Mobile	2020	Strategy	RedFort St	0.95	1.35	0.8	0.9
26	25	Monsoon	Console	2024	Racing	ThunderCl	1.35	1.7	0.95	1.25
27	26	Spice Mar	PC	2022	Simulation	Masala Ga	1	1.45	0.85	1.05
28	27	Indian Oce	Console	2023	Action	BlueWave	1.2	1.6	0.9	1.1
29	28	City Metro	PC	2021	Strategy	UrbanGrid	0.9	1.2	0.7	0.95
30	29	Chai Gully	Mobile	2023	Casual	TasteofL	0.75	1.15	0.65	0.8

SalesData





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▼ Call the active worksheet

```
[4]: ws = wb.active
```

```
[5]: type(ws)
```

```
[5]: openpyxl.worksheet.worksheet.Worksheet
```

```
[6]: ws
```

```
[6]: <Worksheet "SalesData">
```

Call the worksheet with the name

```
[7]: ws=wb['SalesData']
```

```
[8]: ws
```

```
[8]: <Worksheet "SalesData">
```



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Excel

- Count the number of rows and columns in this worksheet

```
] : print(f'Total number of rows: {ws.max_row} and total number of columns: {ws.max_column}')
```

Total number of rows: 31 and total number of columns: 10

Reading data from a cell

```
] : print(f"The value in cell A1 is: {ws['A1'].value}")
```

The value in cell A1 is: Rank

Reading data from multiple cells

```
values = [ws.cell(row=1,column=i).value for i in range(1,ws.max_column+1)]  
print(values)
```

['Rank', 'Name', 'Platform', 'Year', 'Genre', 'Publisher', 'North_Sales', 'South_Sales', 'East_Sales', 'West_Sales']



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Excel

```
data=[ws.cell(row=i,column=2).value for i in range(2,12)]  
print(data)
```

```
['Cricket Champions 2024', 'Delhi Metro Racer', 'Mumbai Crime Stories', 'Temple Quest Legends', 'Royal Chess Masters', 'Battle of Kingdoms',  
Volleyball', 'Hyderabad Tech Tycoon', 'Rajasthan Desert Rally', 'Bengal Mystery Manor']
```

Reading data from a range of cells (from column 1 to 4)

```
my_list = list()  
  
for value in ws.iter_rows(  
    min_row=1, max_row=11, min_col=1, max_col=4,  
    values_only=True):  
    my_list.append(value)  
  
for ele1,ele2,ele3,ele4 in my_list:  
    (print ("{:<35}{:~<8}{:~<10}{:~<10}".format(ele1,ele2,ele3,ele4)))  
    '''Width of first column is 35, second column is 10, third column is 10, fourth column is 10''')
```

Rank	Name	Platform	Year
1	Cricket Champions 2024	PC	2024
2	Delhi Metro Racer	Mobile	2023
3	Mumbai Crime Stories	PC	2022
4	Temple Quest Legends	Mobile	2021
5	Royal Chess Masters	PC	2020
6	Battle of Kingdoms	Console	2023
7	Goa Beach Volleyball	Mobile	2022
8	Hyderabad Tech Tycoon	PC	2021
9	Rajasthan Desert Rally	Console	2020
10	Bengal Mystery Manor	PC	2024



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Excel

- Writing to a cell

```
ws['K1'] = 'Sum of Sales'
```

Specify the row and column position of the cell that has to be written to

```
ws.cell(row=1, column=11, value = 'Sum of Sales')
```

Save the changes since they will not be reflected in the worksheet

Workbook should be closed

```
wb.save('VideoSales_1.xlsx')
```



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Excel

Creating a new column

```
row_position = 2
col_position = 7

total_sales = ((ws.cell(row=row_position, column=col_position).value)+
               (ws.cell(row=row_position, column=col_position+1).value)+
               (ws.cell(row=row_position, column=col_position+2).value)+
               (ws.cell(row=row_position, column=col_position+3).value))

ws.cell(row=2, column=11).value=total_sales
wb.save('VideoSales_1.xlsx')
```

- Code sets the row and column positions of the starting cell for the sales figures.
- The row_position variable is set to 2, which means the sales figures are in the second row of the spreadsheet.
- The col_position variable is set to 7, which means the sales figures start in the seventh column of the spreadsheet.



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Excel

- The `ws.cell()` method is used to access each cell, with the row and column parameters specifying the position of each cell.

```
row_position = 1

for i in range(1, ws.max_row):

    row_position += 1
    NA_Sales = ws.cell(row=row_position, column=7).value
    EU_Sales = ws.cell(row=row_position, column=8).value
    JP_Sales = ws.cell(row=row_position, column=9).value
    Other_Sales = ws.cell(row=row_position, column=10).value

    total_sales = (NA_Sales + EU_Sales + JP_Sales + Other_Sales)
    ws.cell(row=row_position, column=11).value = total_sales

wb.save("VideoSales_1.xlsx")
```



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Add a new row

```
new_row = (  
    31,  
    'Cricket Premier League',  
    'PC',  
    2024,  
    'Sports',  
    'GameStudio India',  
    2.50,  
    1.80,  
    0.90,  
    1.20,  
    6.40  
)  
  
ws.append(new_row)  
wb.save("VideoSales_1.xlsx")
```

Check if the data is written

```
values = [ws.cell(row=ws.max_row,column=i).value for i in range(1,ws.max_column+1)]  
print(values)
```

```
[31, 'Cricket Premier League', 'PC', 2024, 'Sports', 'GameStudio India', 2.5, 1.8, 0.9, 1.2, 6.4]
```



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Excel

Delete a row

```
ws.delete_rows(ws.max_row, 1)  
wb.save('VideoSales_2.xlsx')
```

Excel Formulas

Assigns a formula to cell P2 using the AVERAGE function in Excel.

The formula calculates the average of the values in cells K2 to K31

```
ws['P1'] = 'Average Sales'  
ws['P2'] = '= AVERAGE(K2:K31)'  
wb.save('VideoSales_Formula.xlsx')
```

COUNTA function in Excel counts cells that are populated within a specific range

```
ws['Q1'] = "Number of Populated Cells"  
ws['Q2'] = '=COUNTA(E2:E40)'  
wb.save('VideoSales_Formula.xlsx')
```



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Excel

COUNTIF() counts the number of cells that meet a specific condition

```
ws['R1'] = 'Number of Rows with Sports Genre'  
ws['R2'] = '=COUNTIF(E2:E31, "Sports")'  
wb.save('VideoSales_Formula.xlsx')
```

SUMIF()

Find the total “Sum of Sales” generated by sports games

```
ws['S1'] = 'Total Sports Sales'  
ws['S2'] = '=SUMIF(E2:E31, "Sports", K2:K31)'  
wb.save('VideoSales_Formula.xlsx')
```



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Excel

CEILING()

Rounds a number up to the nearest specified multiple (here 25)

```
ws['T1'] = 'Rounded Sum of Sports Sales'  
ws['T2'] = '=CEILING(S2,25)'  
wb.save('VideoSales_Formula.xlsx')
```

Print the name of the active sheet

```
print(ws.title)
```

SalesData

Rename the worksheet

```
ws.title = 'Games Sales Data'  
wb.save('VideoSales_Formula.xlsx')
```

```
print(ws.title)
```

Games Sales Data



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Excel

Create a new worksheet

```
wb.create_sheet('Empty Sheet')  
print(wb.sheetnames)  
wb.save('VideoSales_Formula.xlsx')
```

```
['Games Sales Data', 'Empty Sheet']
```

Deleting a worksheet

```
wb.remove(wb['Empty Sheet'])  
print(wb.sheetnames)  
wb.save('VideoSales_Formula.xlsx')
```

```
['Games Sales Data']
```

Duplicate a worksheet

```
wb.copy_worksheet(wb['Games Sales Data'])  
wb.save('VideoSales_Formula_2.xlsx')
```

```
print(wb.sheetnames)
```

```
['Games Sales Data', 'Games Sales Data Copy']
```




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Recap

- Openpyxl
- Usage



THANK YOU

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